Venkata Lakshmi Parimala Pasupuleti

Senior Business Analyst · SQL · Python · Statistical Modeling · Cross-functional Analytics · ETL Pipelines · AWS · Generative AI

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PROFESSIONAL SUMMARY

Senior analyst with 8+ years identifying business challenges, modeling data-driven solutions, and partnering cross-functionally with IT, Operations, and Finance teams to drive measurable impact on high-stakes regulated datasets. At Entrata, translated complex compliance and operational requirements into **20+ production analytics modules** across 120+ Agile sprints, driving **80% reduction in manual operational workload** and **30% SQL performance improvement** across 200+ queries on 1M+ records; built **property performance and eligibility dashboards** surfacing KPIs across 100K+ unit portfolios for operations and leadership. At GMU, delivered **self-service BI dashboards** (Tableau, ggplot2, Quarto) improving stakeholder comprehension **40%** and built multi-source regression models achieving **$R^2 = 0.977$** with automated statistical reporting. Actively applying GenAI to analytics and business intelligence: built **AgentProbe** (LLM-as-judge eval engine) and **Leasel** (agentic compliance pipeline, 93.6% precision). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 – Aug 2025

Research Associate — Business Analysis · Statistical Modeling · Self-Service BI · Cross-functional Stakeholder Reporting

- Partnered with faculty leadership to identify research program challenges — built **self-service BI dashboards** (Tableau, ggplot2, Quarto) translating complex model outputs into decision-ready visualizations; improved stakeholder comprehension **40%** across two cohorts with KPI-driven reporting.
- Built end-to-end data modeling and ETL pipelines (Python, R) on multi-source datasets (WHO + UK Road Safety, 190+ countries) — regression models achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 pipelines surfacing statistical outliers manual review missed.
- Championed generative AI adoption in analytics workflows — automated statistical reporting pipelines (Python, Quarto) cutting manual analysis time; applied hypothesis testing, cross-validation, and PCA across multiple research initiatives.

Entrata
Jan 2021 – Aug 2024

Senior Software Engineer — Business Analysis · Data Analytics · Cross-functional Delivery · AWS Data Infrastructure

- Identified business challenges across 100K+ unit portfolios and translated them into scalable analytics solutions — built **property performance and eligibility dashboards** (occupancy, delinquency, eligibility approval/denial rates, rent burden vs. AMI) with KPIs aligned to operations and leadership stakeholders across IT, Finance, and Compliance teams.
- Drove **80% reduction in manual operational workload** by analyzing process bottlenecks and designing Python + PL/SQL automation across income eligibility verification, AMI threshold calculations, and recertification workflows; delivered **30% SQL performance improvement** across 200+ queries on 1M+ records.
- Migrated all client data to **AWS (S3, RDS, Glue)** with encrypted ETL pipelines — **100% PHI/HIPAA compliance**, zero security incidents; cross-functional delivery across IT, Operations, and Compliance stakeholders sustaining **99.5% uptime**.

Entrata
Jan 2019 – Dec 2020

Software Engineer — Compliance Analytics · Eligibility Modeling · Business Requirements Analysis

- Analyzed and encoded manual income eligibility logic into automated analytics pipelines — AMI threshold calculations, income certification workflows, escalation routing across **millions of tenant certifications**; built audit-ready reports with full data lineage for compliance and Finance stakeholders.

Entrata
Jan 2017 – Dec 2018

Associate Engineer — Business Requirements · SQL · Reporting Systems

- Gathered business requirements and built reporting features and SQL data layers (PHP, JavaScript, PostgreSQL) for property management workflows; wrote and tuned queries across all client database environments supporting 100K+ unit portfolios.

PROJECTS

AgentProbe: GenAI Analytics Testing Toolkit

Python · LangGraph · FastAPI · ChromaDB · LLM-as-Judge · OpenAI · Anthropic · Groq · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Built a multi-agent AI platform with a **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) and multi-provider LLM routing — demonstrates practical generative AI integration into analytics and business intelligence workflows at production scale.

CrashClarity: Large-Scale Risk Analytics (104K Records)

Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages — Live Dashboard

- Mined 104K+ UK accident records — built Random Forest + Logistic Regression ensemble achieving **85%+ accuracy**; deployed interactive self-service dashboard on GitHub Pages surfacing risk insights for non-technical stakeholders.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Statistical modeling, ETL pipelines, business analysis, ML analytics, data visualization, AI systems

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

TECHNICAL SKILLS

Business Analysis & Analytics: Business Requirements Analysis, Cross-functional Stakeholder Reporting, Process Automation, Data-driven Decision Support, KPI Design, Statistical Modeling, Regression Analysis, Hypothesis Testing, A/B Testing, Market Research, Quantitative Analysis, Python, R, SQL

BI & Visualization: Tableau, Power BI, ggplot2, Quarto, matplotlib, Jupyter, Self-Service Dashboards, Stakeholder Reporting, Data Storytelling, Isolation Forest, Random Forest, Logistic Regression, Scikit-learn, Pandas, NumPy

Data Infrastructure & Cloud: AWS (S3, EC2, RDS, Glue), ETL Pipelines, Data Pipelines, Data Modeling, Data Warehousing, PostgreSQL, MySQL, Oracle, SQLite, PHI/HIPAA Compliance, Audit Pipelines, Metadata & Lineage, Docker, CI/CD, Git, Linux

Generative AI & Automation: LangGraph, RAG Pipelines, LLM-as-Judge Evaluation, Multi-Provider LLM Routing, OpenAI API, Anthropic API, FastAPI, Automated Reporting Pipelines